

Name : Priyanshu Digari
Roll Number : 89

Assignment -2 Neural Network and Deep Learning

Q1. Explain how a neural network computes its output.

Ans. A neural network computes output in forward propagation:

Input features are passed to neurons
Each neuron computes:

$$z = w \cdot x + b$$

Apply activation function:

$$a = f(z)$$

Output of one layer becomes input to next
Final layer produces prediction

Q2. Explain Hebbian learning rule with an example

Ans.

Principle: "Neurons that fire together,
wire

together. "Rule:

If two neurons activate simultaneously, their connection weight increases.

$$\Delta w = nxy$$

x: input

y: output

n: learning rate

Example:

If input neuron and output neuron are both active $\rightarrow$ weight strengthens.

Q3. Explain why activation function derivatives are required.

Ans. Used in backpropagation

Helps compute gradients of loss function.
Allows updating weights using gradient descent.

Without derivatives, learning (weight update) is not possible.

Q4. What is the exploding gradient

problem?

Ans. Occurs when gradients become very large

Leads to unstable training and huge weight updates

Common in deep or recurrent networks

Effects:

Model fails to converge

Numerical overflow

Q5 Explain the terms:

(i):- vectorization

- Performing operations on arrays/matrices instead of loops.
- Makes computation faster (especially with libraries like NumPy).

(ii):- activation functions

- Functions applied to neuron output
- Introduce non-linearity.

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Examples:

ReLU

Sigmoid

Tanh

(iii):- batch vs mini-batch gradient descent

Type

Description

Batch GD

Uses entire dataset

Mini-batch GD

Uses small subsets

Stochastic GD

Uses one sample